



QUANTITATIVE METHODS 1 · CHAPTER 1

Data Presentation

Detailed solutions

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1. Statistical vocabulary

A survey asks 240 first-year students about programme, transport mode, satisfaction (1-5), number of absences and commuting time in minutes.

- 1 Identify the population, statistical unit and five variables.
- 2 Classify every variable as nominal, ordinal, discrete or continuous.
- 3 Give one possible value for each variable.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. The population is the complete group addressed by the research question; the unit is one entity on which observations are recorded. A variable is classified from the meaning of its values, not from the way it is coded.

Result. Population: the 240 surveyed first-year students, if the objective is to describe those respondents. Unit: one student. Programme and transport mode are qualitative nominal variables because their categories have no natural order. Satisfaction is qualitative ordinal because 1 to 5 rank levels but gaps are not guaranteed equal. Number of absences is quantitative discrete because it is counted. Commuting time is quantitative continuous because time can, in principle, take any value within an interval.

Interpretation. A numerical code does not automatically make a variable quantitative: satisfaction coded 1-5 remains ordinal.

2. Population, sample and bias

KEDGE wants the average commuting time of all 1,200 first-year students. An online questionnaire is posted at 8 a.m.; 180 volunteers answer before noon.

- 1 Define the target population and sample.
- 2 Is the sample necessarily representative? Identify two plausible biases.
- 3 Propose a better sampling design.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Separate the group about which the school wants a conclusion from the group actually observed. Representativeness depends on the selection mechanism, not simply on sample size.

Result. The target population is all 1,200 first-year students; the sample contains the 180 volunteers. Voluntary response can over-represent highly engaged students, and the morning time window can under-represent students commuting or attending class. A better design is a random sample drawn from the student list, possibly stratified by programme or campus, followed by reminders and a non-response analysis.

Interpretation. A sample of 180 can still be biased. Random selection and coverage matter more than the raw response count.

3. Build a frequency distribution

The numbers of purchases made by 20 customers are: 0, 1, 2, 1, 3, 2, 2, 4, 1, 0, 2, 3, 1, 2, 5, 3, 2, 1, 4, 2.

- 1 Construct counts, relative frequencies, cumulative counts and cumulative frequencies.
- 2 Find the share with at most two purchases.
- 3 Choose an appropriate graph and justify it.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Sort the distinct values, count each occurrence, divide by $n=20$, then accumulate from the smallest value.

x	0	1	2	3	4	5
n_i	2	5	7	3	2	1
f_i	.10	.25	.35	.15	.10	.05
N_i	2	7	14	17	19	20
F_i	.10	.35	.70	.85	.95	1.00

Result. The proportion with at most two purchases is $F(2)=14/20=.70=70\%$. A separated bar chart is suitable because the variable is discrete; a cumulative step graph is useful for “at most” questions.

Interpretation. Seven customers, or 35%, made exactly two purchases, whereas 70% made two or fewer.

4. Unequal classes and responsible graphics

Delivery times are grouped as follows: [0,5[: 20; [5,10[: 30; [10,20[: 40; [20,40[: 20.

- 1 Calculate class widths, relative frequencies and frequency densities.
- 2 Explain why plotting counts as bar heights would be misleading.
- 3 Identify the modal class from densities.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. With unequal intervals, histogram area, not height, must represent frequency. Count density is n_i/a_i , where a_i is class width.

Class	[0,5[	[5,10[	[10,20[	[20,40[
Width	5	5	10	20
Count	20	30	40	20
Relative frequency	.182	.273	.364	.182
Count density	4	6	4	1

Result. The modal class by density is [5,10[. Using raw counts as heights would exaggerate wide classes: the [10,20[rectangle would have twice the width of the preceding one and an area unrelated to its frequency.

Interpretation. A correct histogram preserves frequency through rectangle areas and therefore permits fair comparison across unequal classes.

5. Dashboard audit

A manager reports that satisfaction “doubled” because a truncated-axis chart rises from 3.8 to 4.2, and claims the sample represents the school although only members of one student association answered. List four statistical problems, redesign the communication and write a defensible conclusion.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Audit the sampling frame, scale, numerical claim and scope of inference.

Result. First, 4.2 is not twice 3.8; the increase is $(4.2/3.8-1) \times 100 = 10.53\%$. Second, a truncated axis visually magnifies the change. Third, association members form a convenience sample. Fourth, an ordinal average should be accompanied by the full distribution. Use a clearly labelled scale from 1 to 5, show category percentages, disclose n and the recruitment method, and avoid generalising beyond respondents.

Interpretation. A defensible statement is: “Among responding association members, mean reported satisfaction rose from 3.8 to 4.2; this 0.4-point increase cannot automatically be generalised to all students.”

6. Choosing the right chart

For each variable, choose a suitable graph and justify it: study programme; satisfaction from 1 to 5; monthly income; daily sales over 30 days; market shares of four brands.

Method prompt. Match the graph to the variable type and to the analytical question.

Detailed solution

Programme: bar chart. Satisfaction: ordered bars. Income: histogram or boxplot. Daily sales: time-series line chart. Market shares: bar chart is best for precise comparisons; a pie chart is acceptable only if the four shares form one total.

Key point. A graph is selected for a purpose, not for decoration.

7. Cross-tabulation and conditional percentages

Among 120 students, 50 use public transport, including 30 women. Of the 70 other students, 35 are women.

- 1 Build the gender $\times$ transport table.
- 2 Compute the public-transport rate within each gender.
- 3 State whether an association appears present.

Method prompt. Conditional percentages use the total of the conditioning group as denominator.

Detailed solution

The table is: women 30 public and 35 other; men 20 public and 35 other. Totals are 65 women and 55 men. Conditional rates are $30/65 = 46.2\%$ for women and $20/55 = 36.4\%$ for men.

The sample shows a 9.8-point difference. This is descriptive evidence of association, not proof of a population difference or causal effect.

8. Data-quality and communication protocol

A CRM extract contains duplicate customers, missing ages coded 999, inconsistent dates and revenue recorded in euros and thousands of euros. Propose a reproducible six-step protocol before producing a management dashboard.

Method prompt. Separate validation, cleaning, documentation and communication.

Detailed solution

- 1 Preserve an immutable raw file and identifiers.
- 2 Define a data dictionary and units.
- 3 Detect duplicates and specify a defensible retention rule.
- 4 Recode 999 as missing and report missingness by group.
- 5 Validate dates, ranges and logical consistency.
- 6 Standardise monetary units, document every transformation and publish checks with the dashboard.

Never silently replace missing observations or delete unusual values.

9. Continuous-assessment case: willingness to pay

Case study — 12 marks. A food retailer asks 200 customers for the maximum price they would pay for a sustainable lunch box.

Maximum price (€)	[20,25[	[25,30[	[30,40[	[40,50[	[50,60[	[60,80[
Customers	10	20	50	60	40	20

- 1 Identify the population, statistical unit, variable and variable type. (2 marks)
- 2 Calculate class widths, relative frequencies and frequency densities. (3 marks)
- 3 Explain how to construct the correct histogram and identify the modal class. (2 marks)
- 4 Calculate ascending and descending cumulative counts. Give the coordinates needed to draw both cumulative curves. (3 marks)
- 5 Find the highest price that at least 60% of surveyed customers would be willing to pay. Explain the percentile used. (2 marks)

Method prompt. Unequal-width classes require frequency density. For the pricing decision, translate “at least 60% can pay” into an ascending cumulative percentage.

Detailed solution and marking logic

1. Population: the 200 surveyed customers; unit: one surveyed customer; variable: maximum willingness to pay in euros; type: quantitative continuous.

Class	Width	Count	Frequency	Density (count/width)	Ascending count
[20,25[	5	10	5%	2	10
[25,30[	5	20	10%	4	30
[30,40[	10	50	25%	5	80
[40,50[	10	60	30%	6	140
[50,60[	10	40	20%	4	180
[60,80[	20	20	10%	1	200

2–3. Because widths differ, histogram heights must be densities, not raw counts. The largest density is 6, so the modal class is [40,50[.

4. The ascending curve passes through (20,0), (25,10), (30,30), (40,80), (50,140), (60,180), (80,200). The descending curve uses (20,200), (25,190), (30,170), (40,120), (50,60), (60,20), (80,0).

5. If at least 60% must be able to pay, no more than 40% may have a willingness to pay below the chosen price. Forty per cent of 200 is 80, reached at €40. The decision uses the 40th percentile, not the 60th: the recommended maximum price is therefore €40.

Exam habit. State the rule, show the calculation, then interpret the result in the context of the decision.

10. Exam-style synthesis: reporting customer satisfaction

Independent case — 10 marks. Two stores collect satisfaction ratings on an ordered scale from 1 (very dissatisfied) to 5 (very satisfied).

Rating	1	2	3	4	5	Total
Store A	6	10	18	36	30	100
Store B	4	12	30	34	20	100

- 1 Identify the variable and its measurement scale. (1 mark)
- 2 Choose and justify one graph for comparing the two stores. Describe precisely how it should be constructed. (2 marks)
- 3 Compute, for each store, the percentages rated 4 or 5 and the cumulative percentage rated at most 2. (3 marks)
- 4 A manager says: “Store A is 50% better because it has 30 ratings of 5 versus 20.” Criticise this claim. (2 marks)
- 5 Write a three-sentence statistical conclusion, including one limitation of the survey. (2 marks)

Method prompt. Keep the order of the categories, compare percentages because the store totals could differ in another dataset, and separate description from causation.

Detailed solution

1. Satisfaction is qualitative ordinal: categories have a meaningful order, but the distances between adjacent ratings are not guaranteed equal.
2. Use grouped bars or a 100% stacked ordered bar chart. Keep ratings 1–5 in order, use identical axes and colours, and show percentages. A histogram is inappropriate because the categories are ordinal, not continuous intervals.
3. Store A: ratings 4–5 = $(36+30)/100 = 66\%$; ratings at most 2 = $(6+10)/100 = 16\%$. Store B: ratings 4–5 = $(34+20)/100 = 54\%$; ratings at most 2 = $(4+12)/100 = 16\%$.
4. Thirty is indeed 50% greater than twenty, but this isolates one category and does not mean that the store is “50% better”. Across ratings 4–5, the difference is 12 percentage points; the sampling method and uncertainty are unknown, and no causal conclusion follows.

5. Model conclusion. “In the surveyed samples, Store A has a higher share of favourable ratings (66%) than Store B (54%), while both have 16% ratings of 1 or 2. The difference is descriptive and does not by itself establish that Store A provides better service. Before generalising, we need to know how customers were selected, the response rates and whether the survey periods were comparable.”